MILITARY SPECIFICATION

DYNAMOMETER, ENGINE RUN-IN, 500 HORSEPOWER CAPACITY

This specification is approved for use by the US Army Armament Research and Development Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a dynamometer designed and constructed to facilitate the testing and run-in of reciprocating-piston, internal combustion engines having outputs up to and including 500 brake horsepower (Hp).

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

O-A-548	- Antifreeze/Coolant, Engine, Ethylene Glycol, Inhibited, Concentrated
PPP-B-636	- Box, Shipping, Fiberboard
PPP-C-843	- Cushioning Material, Cellulosic
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-60	- Tape, Packaging, Waterproof

MILITARY

MIL-C-104	- Crate, Wood, Lumber and Plywood, Sheathed, Nailed and Bolted
MIL-P-116	- Preservation, Methods of
MIL-T-152	- Treatment, Moisture and Fungus Resistant, Of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	- Varnish, Moisture-And-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-C-5501	- Cap and Plug, Protective, Dust and Moisture Seal, General Specification for
MIL-S-12514	- Starter, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 Hp, Manual and Electric

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards For Federal Services

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of US Military Property
MIL-STD-889	- Dissimilar Metals
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained by the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

ANSI MC96.1 - Temperature Measurement Thermocouples

(Application for copies should be addressed to the American National Standards Institute, Inc, 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

MG-1 - Motors and Generators

Publication WC3 - Rubber Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy
NEMA Electrical Standards

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2101 L Street, NW, Washington, DC 20037.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informal services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified in the contract, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1). The first article(s) may be either standard commercial production model(s), preproduction model(s), or initial production item(s) which conform to the requirements of this specification. In either case, the approved first article(s) and the production items shall be in accordance with the terms of this contract.

3.1.1 Certification. Certification, supported by test data/reports, that the dynamometer is capable of absorbing power as described on the power absorption curve (see 3.4.1.1) shall be furnished with the first article or test report, as applicable. The power curve shall also be included in any technical publications furnished with each dynamometer.

3.2 Design. The dynamometer shall be designed for the mounting, testing, and run-in of reciprocating piston, spark ignition and compression ignition, internal combustion engines up to and including 500 horsepower at speeds from 400 to 5,000 revolutions per minute (rpm). The dynamometer shall be capable of testing and run-in of the following engines: Continental R-6602, GMC 302, Chrysler 75M, Cummins NH-250, Detroit Diesel 8V-71-T, Continental LDS427-2 and LDS465-1, Cummins V8-300, Detroit Diesel 3-53 and 6V-53T, Mack Endt-673, Continental A01-402-5, Continental AOS-895-3 and AOSI-895-5, Continental AO-895-4, and IH DT-1-817. The engine dynamometer shall be so designed that the engines may be readily installed with only

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

minor adjustment for the particular engine being tested. The dynamometer shall include all the equipment necessary, including engine adapters, for the mounting and proper operation of the above engines, except for the following, which (if required) shall be furnished by the Government:

(a) Coolant supply
(b) Coolant disposal
(c) Compressed air supply
(d) Electric power supply
(e) Engine fuel supply
(f) Engine air inlet
(g) Engine exhaust disposal
(h) Engine hoisting equipment
(i) Fire extinguishing equipment
(j) Instruments for measuring atmospheric conditions
(k) Storage batteries

3.2.1 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking lubricant levels. The supplier shall include and install all lubricants, fluids, greases and other compounds in the dynamometer, in accordance with the best commercial practices of the dynamometer industry.

3.2.2 Safety provisions. The dynamometer shall incorporate safety devices to prevent conditions hazardous to personnel or detrimental to equipment (see 3.2.2.1, 3.4.3, and 3.4.4.4). Additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.3).

3.2.2.1 Overspeed trip governor. An engine overspeed trip governor shall be furnished to provide overspeed protection for both spark ignition and compression ignition engines. The overspeed governor shall override the throttle control, and either stop the engine completely or shall return the engine to a fast idle, in the event the engine speed exceeds the governor setting. The overspeed governor shall be graduated in increments not greater than 100 rpm, and shall be adjustable through a 2,000 to 5,000 rpm engine speed range. The governor speed selector shall be panel mounted on the control console specified in 3.4.4.

3.2.3 Interchangeability. All replaceable parts shall be designed and manufactured to such standards as shall permit replacement and adjustment without modification of the parts or the dynamometer.

3.3 Construction.

3.3.1 Materials. Materials not specifically designated herein shall be selected by the manufacturer and shall be subject to all requirements specified herein. The manufacturer is encouraged to use reclaimed raw materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected and specified. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

3.3.1.1 Dissimilar metals. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.2 Castings and forgings. All castings shall be free from blow-holes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability.

3.3.3 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where these operations are a requirement of the original design and shall not be used for the repair of defective parts.

3.3.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3.6 Surfaces. All surfaces of the dynamometer shall be clean and free from any harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with standard commercial practice in the industry. Before painting, all surfaces shall be clean and free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.4 Product description. The dynamometer shall consist of the following major components as specified in 3.4.1 through 3.4.6.

3.4.1 Power absorption system. The power absorption system shall include a power absorber, power absorber frame, temperature controls, load controls, throttle control, engine fuel system, engine electrical system, and junction box as specified in 3.4.1.1 through 3.4.1.8.

3.4.1.1 Power absorber. The power absorber (and complete dynamometer) shall be capable of continuously absorbing the horsepower values at the revolutions per minute (rpm) given below. All values shall be considered to be observed values:

20Hp at 600 rpm ± 50 rpm
160Hp at 1,200 rpm ± 100 rpm
335Hp at 1,800 rpm ± 100 rpm
500Hp at 2,300 rpm ± 100 rpm
200Hp at 5,000 rpm ± 100 rpm

The minimum controllable Hp curve shall not exceed the following:

10Hp at 2,000 rpm ± 100 rpm
140Hp at 5,000 ± 100 rpm

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

The power absorber shall be of the hydraulic, eddy current, or electric generator type and shall be bi-directional rotation. Connections to the power absorber shall be such that they cannot affect power measurements. The power absorber shaft bearing seals for water cooled and hydraulic types shall be capable of maintaining a seal under normal operating conditions without requiring external cooling. The seals shall prevent cooling water and lubricant from leaking out and foreign material from entering. Seal replacement shall not require dynamometer major disassembly. The power absorber shaft bearings shall be of the antifriction type. Seals shall be provided to prevent fluid from entering the bearings when the power absorber is of the water cooled or hydraulic type. Load control of the power absorber shall be in accordance with 3.4.1.4. Temperature control of the power absorer, if required for proper dynamometer operation, shall be in accordance with 3.4.1.3.

3.4.1.2 Power absorber frame. The power absorber shall be mounted in the power absorber frame. The bearings supporting the power absorber shall be flexure supports and/or antifriction type and designed for at least 20,000 hours of operation without failure or degradation of dynamometer accuracy due to bearing race indentation under static vibratory load. The frame shall provide support for the power absorbed and related equipment for all specified dynamometer operational requirements.

3.4.1.3 Dynamometer cooling temperature controls. Automatic temperature controls shall be adequate for maximum rated power absorption to prevent the absorption or cooling fluid from reaching its boiling point. The system shall provide for manual control in the event of automatic control failure. In the case of a closed cooling system, the thermostat shall control the flow of cooling fluid through the heat exchanger. The dynamometer power absorption unit water consumption shall not exceed 4.5 gallons per Hp per hour. A low water pressure safety switch shall be connected to the trip governor control (see 3.2.2.1) in such a manner as to reduce engine speed to a fast idle in the event of water pressure failure. A replaceable screen of 40 mesh shall be incorporated in the fluid line to screen out particles of rust and other foreign matter. Equipment covered by this paragraph shall be capable of successful operation with a water supply pressure ranging from 35 pounds per square inch gage (psig) to 70 psig and a water supply temperature ranging from 80 degrees Fahrenheit (F) to 100 degrees F.

3.4.1.4 Load controls. The dynamometer shall be capable of controlling the load on the engine to within $\pm$1 percent of the torque indicator full scale value, and $\pm$1 percent of the full scale horsepower indicator value. The load controls shall provide coarse and fine adjustment of any load within the dynamometer capacity range in no more than 1 minute. The power absorber load controls shall be mounted on the control console specified in 3.4.4.

3.4.1.5 Throttle control. A throttle actuator shall be furnished to provide throttle control for all engines listed in paragraph 3.2. The throttle actuator shall be capable of operating in conjunction with the overspeed trip governor specified in paragraph 3.2.2.1. The throttle actuator shall be installed on the power absorber frame and shall be connected to the engine throttle lever with an adjustable linkage. The throttle actuator shall be controlled by the control transmitter mounted on the control console as specified in para 3.4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

3.4.1.6 Engine fuel system. The engine fuel system, supplied as part of the dynamometer system, shall conduct fuel from a supply outlet, provided by the Government (see 3.2), through a shutoff valve, fuel filters, selector valves, fuel flow measuring units, modulating valves, and fuel lines. The fuel system shall be mounted on a panel which can be easily attached and removed from the power absorber frame. All valves shall be of the packless type, and all metal piping shall be brass or copper or stainless steel. The entire system shall permit passage of 400 pounds of fuel per hour with a pressure drop in the system of less than 4 psig. Convenient and efficient bleed-offs shall be provided to eliminate vapor locks. Filters shall be of the replaceable element type and shall have a rating of not more than 5-microns. Two independent fuel systems, each with a primer pump, a shutoff and pressure modulating valve, and a pressure gage, shall be provided for (1) gasoline fueled carburetor and gasoline fuel injection type engines and, (2) compression-ignition type engines. Primer, pumps, valves, and pressure gages for both systems shall be provided.

3.4.1.7 Engine electrical system. The engine electrical system shall be a self-contained unit which can be easily attached and removed from the power absorber frame. The engine electrical system shall include provisions for two 12 volt vehicle wet storage batteries (wired in series) and an insulated wiring system to permit the operation of engines on negative grounded 12 and 24 volt ignition systems. The engine electrical system shall include provisions for remote control operation of starting relays, and booster coils through the use of an ignition switch mounted on the control console. Starting relays and circuits shall be capable of handling maximum starting currents. All cable used in the starting circuits shall be AWG copper conductor, heavy duty neoprene jacket, portable arc welding cable conforming to NEMA Publication WC3 and shall be capable of transmitting imposed electrical currents and voltages. An automatic battery charger, with a two stage charging rate, shall be equipped with an automatic regulator to provide a charging rate of 5 to 6 amperes per hour when the battery voltage drops below normal, and to provide a "trickle" charge of not more than one ampere per hour to keep the batteries fully charged at all times without the danger of over-charging or overheating the batteries. The charger shall be equipped with a meter and a meter switch to permit visual observation of the charging rate. The two 12 volt storage batteries shall not be furnished with the dynamometer.

3.4.1.8 Junction box. The juction box shall be attached to the power abosrber frame. The junction box shall be for connecting leads to their respective systems and shall be a NEMA 12 electrical enclosure.

3.4.2 Engine stands. Two mobile engine stands shall be provided and shall support engines weighing up to 5,000 pounds. Adjustable front and rear mounts as well as engine mounting adapters shall be provided for the installation and alignment of all engines listed under paragraph 3.2. Each stand shall be mounted on a minimum of four wheels. The wheels on one end shall be the rigid mounted nonswivel type and the remainder of the wheels shall be 360 degree swivel caster type. The wheels shall have a minimum diameter of four inches and shall provide free steerable movement of the loaded engine stand in any direction over the floor surface. The wheels shall be provided with antifriction bearings in the swivels and wheels. Wheel location on the engine stand shall effectively support movement and prevent an unbalanced or tipping condition of the stand when loaded with the maximum weight of 5,000 pounds. The swivel wheels shall be provided with foot actuated brakes which may be applied to prevent movement of the engine stand. Mechanical guides and a locking device shall be provided to permit aligning and locking the mobile engine stands to the power absorber frame. The stand(s) shall be equipped with a drain pan having a tapered pipe thread of sufficient size (3/4 NPT minimum)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

with mating pipe cap. No hard plumbing is proposed.

3.4.3 Engine cooling water column. A thermostatically controlled cooling column shall be furnished for use with liquid cooled engines. The heat dissipation rate shall be at least 1,320,000 BTU per hour. An adjustable thermostatic control valve shall be provided to obtain an engine water outlet temperature of 160 degrees F to 200 degrees F. A pressure relief valve shall be incorporated in the cooling system to prevent abnormal cooling system pressure. A means shall be provided for filling, venting and draining the cooling system. The cooling column shall be a free standing and self-contained design. Hoses with quick disconnect couplings and adapters shall be provided to connect the cooling column with the engines listed in paragraph 3.2. The cooling system shall be equipped with a sight gage to indicate the coolant level and a dial thermometer to indicate the engine coolant temperature.

3.4.4 Control console. The control console shall house all controls and instruments necessary for engine testing. The control console shall incorporate a data desk in a position below and in front of the instrumentation and control boards. Instruments and controls shall be oriented to promote legibility, and ease of operation by the seated operator. The control console shall be free standing and shall be completely enclosed. All internal electrical leads, thermocouple leads and mechanical leads shall be terminated on a junction panel in the rear of the control console. Access from the rear of the control console shall be provided for connecting electrical and mechanical leads and for servicing and repairing all instruments and controls. The control console shall be arranged for installation outside of the test cell. The instruments and controls shall be permanently marked for function, and grouped, as far as possible, in four general areas as specified in 3.4.4.1, 3.4.4.2, 3.4.4.3, and 3.4.4.4. A receptacle shall be furnished and mounted on the control console to house a supply of test-log forms. A supply of 100 test-log forms shall be furnished.

3.4.4.1 Engine controls. The engine controls shall include the panel mounted ignition switch specified in 3.4.1.7, a panel mounted switch for the battery system, and the engine throttle control transmitter. Start-stop controls shall be included for the priming pumps for both fuel systems (see 3.4.1.6).

3.4.4.2 Engine instrumentation. The engine instrumentation shall include the following panel mounted indicators. The indicators shall be capable of receiving, and indicating oil pressure, intake manifold pressure and vacuum, fuel pressure, fuel flow rate, cylinder head temperature, coolant temperature, and oil temperature as noted:

(a) One mechanical engine oil pressure indicating device with a range from 0 to 160 pounds per square inch (psi) graduated in increments not greater than 5 psi, and an accuracy error not greater than $\pm$ three percent of the full scale value from 0 to 40 psi and 120 to 160 psi and $\pm$ two percent of full scale value from 40 to 120 psi.
(b) Two mechanical or one compound fuel pressure indicating devices with one for gasoline and one for diesel fuel, a range from 0 to 15 psig pressure and 0 to 30 inches of mercury vacuum, graduated in increments not greater than .5 psi and 1 inch, and an accuracy error not greater than $\pm$ five percent of full scale value.
(c) One manifold pressure indicating device with a range of 0 to 70 inches of mercury absolute pressure, graduated in increments not greater than .5 inch, and an accuracy error not greater than $\pm$ 1/2 inch of mercury.
(d) One thermocouple type engine temperature indicator having at least eight stations with automatic cold junction temperature compensation, a range of 0

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

to 600 degrees F, and an accuracy error not greater than ± one percent of full scale value. Eight pairs of No. 14 AWG, or larger, Iron-Constantan thermocouple lead wires, special tolerance, insulated, stranded conductor, in accordance with ANSI MC 96.1, shall be used from the instrument to the dynamometer junction panel thermocouple outlets. The instrument shall be compensated to match the length of lead wire from the indicating instrument to the engine thermocouples. Eight Iron-Constantan thermocouples with compression fittings shall be provided to mount in various locations on the engines to be tested.

(e) Two fuel flow indicating devices or one compound fuel flow indicating device (one for spark-ignition fuel, and one for compression-ignition fuel) with an accuracy error not greater than ± two percent of actual value over the specified range shall be furnished. The working range of the device shall be from 15 pounds per hour to 500 pounds per hour. A selector switch shall be provided to permit the use of either spark ignition fuel or compression ignition fuel. The selector switch shall operate in conjunction with a solenoid selector valve furnished with the fuel system in para 3.4.1.6.

3.4.4.3 Dynamometer instrumentation and controls. The dynamometer instrumentation and control area shall include the following panel mounted indicators and shall be capable of receiving, transmitting, and indicating engine speed, torque, and horsepower, as noted. Digital display indicators shall be furnished for (a) and (b) and have an accuracy error not greater than ± one percent of actual value over the specific range and ± one digit. All display devices shall be readable at a distance of five feet by the operator. The speed, torque and horse power indicators shall be equipped with built-in calibration reference for easy front-panel calibration.

(a) One magnetic pulse type engine rpm sensor with leads to an electric tachometer indicating. The digital display indicator shall have a range from 0 to 5,000 rpm.
(b) One torque and one horsepower measuring system with torque sensing strain gage load cells directly actuated by the power absorber torque arm. All necessary leads shall be furnished from the torque sensing device to the torque and hp indicators. The torque indicator shall have a range from 0 to 1200 foot-pounds (minimum). The horsepower indicator shall have a range from 0 to 500 hp.
(c) One coolant supply pressure indicating device with a range of 0 to 100 psi graduated in increments not greater than 5 psi. Accuracy error shall be within ± three percent of full scale value.
(d) One power absorber fluid outlet temperature device (coolant and hydraulic type power absorbers only). Accuracy error shall be within ± three percent of full scale value.
(e) The dynamometer load controls shall be as specified in 3.4.1.4.

3.4.4.4 Safety warning devices. Control console panel mounted individual warning lights or buzzers shall be furnished to indicate the following conditions:

(a) High power absorber fluid temperature (water cooled and hydraulic type power absorbers only).
(b) Low engine oil pressure. The indicator light shall be connected to the proper terminal on the junction panel, located on the junction box specified in 3.4.1.8.
(c) High engine oil temperature. The indicating light shall be connected to the proper terminal on the junction panel,located on the junction box

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

specified in 3.4.1.8.
(d) High engine coolant temperature. The indicator light shall be connected to the proper terminal on the junction panel, located on the junction box specified in 3.4.1.8.
(e) Tripped overspeed governor.

3.4.4.5 Control console leads. All electrical and mechanical leads required for remote engine operation shall be furnished, shall pass through separate conduits, and shall be grouped as follows:

(a) Instrument electrical leads
(b) Engine and operating 24 volt lead
(c) Thermocouple leads
(d) Primary power leads
(e) Mechanical leads

The length of the leads shall be 30 feet (minimum). Electrical leads shall terminate in terminal eyes. Instrument leads shall have soldered connections at both ends. All leads shall be plainly identified at both ends.

3.4.5 Drive shaft. A dynamometer drive coupling shall be provided and shall be capable of transmitting power over the horsepower and speed ranges specified in 3.4.1.1. The drive coupling shall compensate for angular misalignment of at least three degrees between the power absorber shaft and the engine shaft. The drive coupling shall have a lengthwise adjustment of at least 1 1/2 inches.

3.4.6 Adapter kits. Adapter kits shall be provided for testing and run-in of the engines specified in paragraph 3.2 (see 6.2.1). Painting of Adapter Kit components shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: All surfaces to be painted shall be cleaned to be free of all foreign matter detrimental to painting and at least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent one coat epoxy enamel applied.

3.5 Hardware and special tools. All hardware required for the complete and proper installation of the dynamometer shall be furnished. All special tools necessary for the proper installation, operation, calibration, and maintenance of the dynamometer, and instruments shall be furnished.

3.6 Electrical equipment. Electrical components, including starters, relays, switches, wiring, and their enclosures, shall conform to and be located in accordance with NEMA Electrical Standards except as specified herein. No component shall be used in any application where the requirements exceed the manufacturer's rating for the part. Connections to screw type terminals shall be mechanically secured to prevent loss of tightness. Properly crimped or soldered (brazed) terminal lugs are acceptable. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

3.6.1 Motors and starters. Electric motors shall conform to the requirements of NEMA MG-1. Starters shall conform to the requirements of MIL-S-12514.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

3.6.2 Power source. The dynamometer shall be designed for operation on nominal 220/440 volt, 50/60 hertz (Hz), three-phase current. Control switches and circuits shall operate on a nominal 115-volt, 50/60 Hz, single phase current.

3.7 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.8 Product identification. The dynamometer shall be marked for product identification in accordance with MIL-STD-130 and, unless specified (see 6.2.1), shall include the National Stock Number.

3.9 Workmanship. Standards of workmanship shall be such as to assure equipment with characteristics found in standard commercial units and as specified herein. Particular attention shall be given to freedom from blemishes, defects, and sharp edges; thoroughness of soldering, welding, brazing and painting; and alignment of parts and tightness of screws and bolts.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. When specified, first article inspection shall be performed on the preproduction model(s) or initial production item(s) noted in 3.1. First article inspection shall consist of the examination in 4.5, and all tests of 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production dynamometers offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (e) as follows, and failure of any unit to pass an examination or test shall be cause for rejection.

(a) Product examination (see 4.5)
(b) Power absorber test (see 4.6.1)
(c) Safety controls tests (see 4.6.3)
(d) Instrument accuracy test (see 4.6.4)
(e) Packaging inspection (see 4.7)

4.4.1 Sampling. Sampling for quality connformance inspection shall be performed in accordance with MIL-STD-105, using the following:

Inspection or test	Level	AQL
Product Examination (see 4.5)	II	1.0
Power absorber test (see 4.6.1)	II	1.0
Dyno temperature control test (see 4.6.3.1)	II	1.0
Low dyno water pressure test (see 4.6.3.2)	II	1.0
Overspeed trip governor test (see 4.6.3.3)	II	1.0
Instrument accuracy test (see 4.6.4)	S-4	1.0
Packaging Inspection (see 4.7)	S-4	4.0

4.5 Product examination. The dynamometers shall be visually and manually examined to determine conformance with the requirements of 3.2 through 3.2.2, 3.2.3 through 3.4.1, and 3.4.1.2 through 3.9. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the dynamometers and their components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.3), shall be cause for rejection.

4.6 Tests. In the event an engine with the proper capacity cannot be obtained for these tests, it is permissible to utilize a transmission, gears, or other means to increase or reduce speed to obtain the horsepower and speed combinations specified.

4.6.1 Power absorber test. The dynamometer power absorber shall be motored with an applicable power source for a period of not less than 10 minutes at each of the following conditions (see 3.4.1.1):

20 hp at 600 rpm $\pm$ 50 rpm
160 hp at 1200 rpm $\pm$ 100 rpm
335 hp at 1800 rpm $\pm$ 100 rpm
500 hp at 2300 rpm $\pm$ 100 rpm
200 hp at 5000 rpm $\pm$ 100 rpm

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

4.6.2 Engine cooling requirement. The temperature of the engine cooling system shall not exceed that rated by the manufacturer at each of the specified load and speed conditions (see 3.4.3). Exceeding the temperature requirements during any tests shall be cause for rejection of the dynamometer.

4.6.3 Safety controls tests. With the engine operating at 190 to 210 horsepower and 1900 to 2100 rpm, conduct the listed safety controls tests.

4.6.3.1 Dynamometer temperature control test. Gradually reduce cooling water flow to increase cooling water temperature. Failure of the automatic temperature control to reduce engine speed to fast idle before the cooling water temperature at the outlet reaches the boiling point shall be cause for rejection (see 3.4.1.3).

4.6.3.2 Low dynamometer cooling water pressure test. Reduce cooling water inlet pressure. Failure of the low water pressure safety switch to reduce engine speed to fast idle before the cooling water inlet pressure goes below 35 psig shall be cause for rejection (see 3.4.1.3).

4.6.3.3 Overspeed trip governor test. Determine fast idle speed and maximum rated speed of the engine used for this tests. Set the overspeed trip governor to function at maximum rated engine speed to 100 rpm above that speed. Failure of the overspeed trip governor to return the engine to fast idle speed or to shut the engine down when the engine reaches overspeed governor trip setting, shall be cause for rejection (see 3.2.2.1).

4.6.4 Instrument accuracy test. The indicating instruments specified in 3.4.4.2 and 3.4.4.3 shall be tested by comparison with standard measuring devices having an overall accuracy within 1/4 of one percent of the value being measured. Each instrument shall be tested to determine conformance to the applicable requirements. The engine instruments of 3.4.4.2 may be tested for accuracy individually or in conjunction with the dynamometer system operation. The dynamometer and controls of 3.4.4.3 shall be tested in conjunction with the dynamometer system operation. Failure of the instruments to meet their specified accuracies shall be cause for rejection of the dynamometer.

4.6.5 Operational test. The dynamometer shall be prepared for a test run in accordance with the manufacturer's instructions. An internal combustion engine, capable of providing the test conditions listed below shall be furnished by the contractor and used for this test (see 3.2 and 3.4.1.1). Test conditions shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

Test Period	Time (minutes)	Engine Speed (rpm)	Net Observed Brake Hp
1	5	Idle	3 to 5
2	5	1000 ± 100	15
3	5	1600 ± 100	20
4	5	2000 ± 100	10 (Minimum load control check)
5	5	2000 ± 100	60
6	5	Any speed between 2600 and 5000	175
7	5	Any speed between 1500 and 5000	400
8	5	5000 ± 100	140 (Minimum load control check)
9	5	Idle	3 to 5

Failure of any part of the dynamometer or its equipment to meet the applicable performance requirements shall be cause for rejection of the dynamometer.

4.6.6 Endurance test. The dynamometer shall be subjected to a 400 hp load at any continuous speed between 1200 and 5000 rpm for a period of not less than four hours. The temperature of the power absorber shall not exceed that rated by the manufacturer at the specified load and test speed. Failure of the dynamometer to comply with the load control requirements (see 3.4.1.4), excessive engine temperature (see 3.4.3), failure of any dynamometer or equipment parts to meet applicable requirements, or evidence of coolant leakage shall also be cause for rejection of the dynamometer.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging the dynamometer to determine compliance with the requirement of Section 5.

5 PACKAGING

5.1 Disassembly. When practical, the test stand/dynamometer shall be disassembled to afford protection of components, attachment, and accessories against damage and pilferage and to reduce cubage. Disassembled parts shall be preserved, anchored, racked, blocked, and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked to identify the proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special tools or skills.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

5.2 Preservation. Preservation shall be Level A or industrial as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. the dynamometer shall be cleaned in accordance with process C-1 and dried in accordance with procedures D-1 through D-5 of MIL-P-116.

5.2.1.2 Unit protection. The unit method of packaging and preservatives specified herein shall conform to MIL-P-116.

5.2.1.2.1 Electrical accessories and components. The control(s) and instrumentation panel, including all electrical accessories and components such as cables, motors, starters, junction boxes and electrical leads, shall be packaged as a unit in accordance with the applicable sub-method of method II of MIL-P-116. If the accessories and components are not disassembled from the unit they should be sealed with barrier material and tape conforming to PPP-T-60 type IV.

5.2.1.2.2 Technical data. Technical data shall be packaged in accordance with method IC-I of MIL-P-116.

5.2.1.2.3 Power absorption unit. All exposed bare metal areas of the power absorption unit shall be coated with type P-11 preservative, and shall be packaged in accordance with method I. The power absorption unit may be packaged in accordance with method III when the absorption unit and frame are completely painted.

5.2.1.2.4 Oil and fuel systems. In accordance with para 3.2.1, this document, all lubrication systems shall contain the supplier's lubricants. However, in the event the lubricant cannot be left in the machine, the oil and fuel systems, including storage tanks, shall be flushed with type P-10, grade 30 preservative. The systems shall be thoroughly drained. After draining, seal the inlets and outlets with tape conforming to PPP-T-60, type IV, or plastic caps or plugs conforming to MIL-C-5501. In the event a lubricant reservoir cannot be covered and sealed, the appropriate lubricant shall be provided in a sealed container and shipped with each dynamometer. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to install the furnished lubricant before operation of the equipment. The operator's manual shall also reflect the above warning and procedure. Vent openings in the lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps or plugs intended to be vented to the atmosphere. These warnings shall also be included in the instruction manual.

5.2.1.2.5 Water system. The complete water system shall be flushed with a solution consisting of 50 percent antifreeze (ethylene glycol), conforming to O-A-548, type I, and 50 percent clean water. The system shall be thoroughly drained after flushing. All openings to the system shall be sealed with plastic cap(s) or plugs conforming to MIL-C-5501 or tape conforming to PPP-T-60, type IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

5.2.1.2.6 Bearings. All bearings used throughout the test stand and all revolving shafts shall be preserved (except sealed for life) with type P-11 preservative and wrapped with barrier material.

5.2.1.2.7 Assembly components. All assembly components such as engine stand, drive shaft, adapter set, pump assemblies, fastening devices (nuts, bolts, pins, washers, and screws), pipe, pipe fittings, and other hardware, including special tools, shall be packaged in accordance with method I of MIL-P-116. In the event the components are completely painted, plated, or fabricated of stainless steel, no preservative is required. When practicable, components shall be placed in fiberboard boxes conforming to PPP-B-636, class weather-resistant. Parts vulnerable to damage shall be cushioned with material conforming to PPP-C-843, type II.

5.2.1.2.8 Hose. All hose issued with the dynamometer shall be coiled to a safe diameter and secured with not less than three pieces of cotton twine or tape evenly spaced about the coil and placed in the assembly boxes or secured to the machine in a safe place.

5.2.1.2.9 Gages and safety warning lights. Gages and safety warning lights shall be covered with a one inch thick uncompressed cushioning material conforming to PPP-C-843, type II, or fiberboard conforming to PPP-F-320, class weather-resistant. Secure the material in place with tape conforming to PPP-T-60, type optional.

5.2.2 Industrial. Industrial preservation shall be as specified in ASTM D3951.

5.3 Packing. Packing shall be Level A or industrial, as specified (see 6.2.1).

5.3.1 Level A. The packaged control console complete with instrument panel and all electrical accessories shall be packed in a crate conforming to MIL-C-104. All other component parts packaged shall be packed in one or more crates conforming either to MIL-C-104. Blocking, bracing, and anchoring shall conform to the crate specification and MIL-STD-1186. Closure and strapping shall conform to the crate specification.

5.3.2 Industrial. Industrial packing shall be as specified in ASTM D3951.

5.4 Marking. Marking shall be Military or industrial, as specified (see 6.2.1).

5.4.1 Military. Military marking shall be as specified in MIL-STD-129.

5.4.2 Industrial. Industrial marking shall be as specified in ASTM D3951.

6 NOTES

6.1 Intended use. The dynamometer covered by this specification is intended to be used for testing and run-in of new or rebuilt reciprocating-piston, internal combusion engines prior to the release of the engines for use in vehicles or for forwarding to supply channels or storage facilities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article, when required (see 3.1).
(c) Additional safety and health requirements (see 3.2.2 and 6.3).
(d) Painting, if different (see 3.3.7).
(e) Engine adapter kits required (see 3.4.6).
(f) Product identification marking, if different (see 3.8).
(g) Selection of level of preservation, packing, and marking required (see 5.2, 5.3, and 5.4).

6.2.2 Contract data requirements. Required technical data such a operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.2.3 Repair parts. Repair part identification in repair parts manuals shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification in repair part manuals shall also include the original equipment/part manufacturer's (OEM) (actual part manufacturer) name, Federal Supply Code for Manufacturer's (FSCM), nomenclature, part number, and price of the part used in the end item immediately after the prime contractors part identification. Item with high mortality rates shall be so identified (see MIL-M-7298).

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and controls of noise levels, radiation, electromagnetic emissions, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Subject term (key word) listing.

Adapter kits
Control console
Dynamometer
Engine stands
Overspeed
Power absorber

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
Army - AL

Review activities:
Army - ME

User Activities:
Army - AT

Preparing Activity
Army - AL

Project Number 4910-A641

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armement Research and Development Command
ATTN: DRDAR-TST-S
Dover, NJ 07801

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.